

Hybrid NLP Framework for Temporal Emotion Trajectory Modeling in Clinical and Informal Text

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Abstract

Depression remains one of the most prevalent mental health disorders worldwide, yet scalable and objective tools for early detection are limited. This work introduces a **hybrid natural language processing (NLP) framework** that integrates transformer-based contextual embeddings with handcrafted linguistic features, applied to clinical notes and patient-generated text. Evaluated on MIMIC-III and CLPsych datasets, the framework achieved **96% accuracy**, outperforming standalone transformer baselines including DeBERTa (0.92), BioClinicalBERT (0.90), and Longformer (0.91).

Interpretability was central to the design: SHAP analyses highlighted sentiment polarity, syntactic drift, and discourse coherence as dominant features, while local force plots provided case-level explanations. These provisions balance predictive accuracy with transparency, ensuring clinically meaningful outputs. Findings confirm that hybrid architectures combining deep embeddings with interpretable features advance mental health risk detection. Future work will extend to multilingual and multimodal sources, transformer-based temporal models, and longitudinal clinical validation.

Index Terms

Natural language processing (NLP), depression detection, clinical notes, social media analysis, transformer embeddings, handcrafted linguistic features, interpretability, SHAP, emotion trajectory, syntactic drift, hybrid classifier, mental health monitoring, early intervention, risk modeling.

I. Introduction

Mental health monitoring increasingly relies on natural language processing (NLP) to detect early risk signals in patient communication. Traditional models often analyze isolated text samples, which means they miss longitudinal patterns that are critical for early intervention. To address this limitation, we present a hybrid NLP framework that models emotion trajectories and syntactic drift across both clinical notes and patient-generated messages. The framework is

designed to achieve high accuracy while remaining interpretable, enabling clinicians and researchers to understand not only what the model predicts but also why those predictions are made.

Rather than treating patient messages or clinical notes as static snapshots, this study focuses on tracking emotional and linguistic changes longitudinally. The goal is to detect gradual shifts in tone, coherence, and sentiment that may signal the onset of depression or anxiety. This approach leverages multimodal data—including clinical notes, therapy transcripts, and patient-generated messages—and applies NLP models to construct emotion trajectories and discourse evolution graphs.

Key innovations of this work include:

- **Temporal modeling** with sliding windows and time-aware embeddings.
- **Emotion trajectory graphs** to visualize polarity and intensity shifts.
- **Discourse coherence tracking** to measure syntactic stability.
- **Hybrid fusion** combining transformer embeddings with handcrafted linguistic features.

By introducing temporal depth, our framework enables systems to identify subtle shifts in emotional and linguistic patterns—a critical capability for early intervention. This design bridges the gap between clinical relevance and linguistic modeling, making it suitable for real-world deployment in healthcare settings.

Timely identification of depression has numerous advantages, greatly enhancing the likelihood of successful therapy and recovery. Early detection can prevent escalation into more severe conditions and ensure that individuals receive appropriate support at the earliest stages. Techniques for identifying depression have included analyzing clinical data or electronic health records [3,4], monitoring personal disclosures on social media [5,6], and examining speech patterns [7,8]. This study emphasizes social media data, as platforms such as Twitter and Reddit have become primary spaces for emotional expression. By performing a comparative examination of transformer-based methods, we aim to obtain a thorough understanding of the contextual factors associated with mental health challenges as expressed in online discourse.

Beyond its technical advances, this work also considers the clinical impact of AI-enabled monitoring for depression and suicide risk. When online behavior reveals signals of suicidal ideation or severe depressive states, automated systems could support timely intervention by notifying mental health professionals, presenting crisis resources, or linking individuals to digital care platforms. Given that social media increasingly functions as a space for emotional disclosure, harnessing AI for early detection offers a pathway to strengthen suicide prevention and improve mental health services [14]. At the same time, issues of privacy, informed consent, and the risk of misclassification must be carefully managed to ensure that such technologies are deployed responsibly in clinical practice [9].

This paper is organized into five sections. Section I introduces the study. Section II reviews previous research in NLP for mental health detection. Section III outlines the research methodology, including datasets, feature engineering, and model architecture. Section IV

presents experiments and results, including classification performance, interpretability analysis, and case studies. Section V concludes with implications, limitations, and directions for future work.

II. Literature Review

Recent research has increasingly focused on leveraging natural language processing (NLP) for mental health detection, with particular emphasis on clinical notes, social media data, and multimodal signals. This section reviews key contributions from 2023–2025 that inform the design of our hybrid framework.

A. Clinical Data and Electronic Health Records

Studies using clinical notes and electronic health records (EHRs) have demonstrated the potential of NLP for depression severity classification. Ahmed et al. [3] introduced an interpretable NLP framework that integrates transformer embeddings with clinical features, achieving strong performance while maintaining transparency for clinicians. Tian et al. [4] extended this line of work by incorporating EEG sensor data into multimodal fusion models, highlighting the value of combining textual and physiological signals for more accurate diagnosis. These approaches underscore the importance of interpretability and multimodal integration in clinical contexts.

B. Social Media Disclosures

Social media platforms such as Twitter and Reddit have become primary outlets for emotional expression, making them valuable sources for early risk detection. Lan et al. [5] explored the use of large language models (LLMs) for depression detection on social media, demonstrating improved contextual understanding compared to traditional methods. Hameed et al. [6] emphasized explainability by applying SHAP and LIME to social media-based depression detection, ensuring that predictions are interpretable and actionable. Together, these studies highlight the dual need for accuracy and transparency in analyzing patient-generated text.

C. Speech and Multimodal Signals

Beyond text, speech patterns provide critical cues for detecting depression and suicidal tendencies. Baydili and Kaya [7] proposed a hybrid deep learning approach that integrates text and speech features, achieving robust performance in suicidal tendency detection. Hasan et al. [8] conducted a comparative study of transformer and LSTM architectures for multimodal depression detection, demonstrating that transformer-based models outperform traditional recurrent networks in capturing complex emotional signals. These findings reinforce the value of multimodal approaches that extend beyond text alone.

D. Gaps in Existing Work

While these studies have advanced the field, several limitations remain. Most existing models either focus on static text samples or single modalities, overlooking longitudinal changes in

emotional tone and syntactic coherence. Few approaches explicitly model temporal emotion trajectories or discourse drift, which are critical for early detection of mental health deterioration. Moreover, balancing predictive performance with interpretability continues to be a challenge, particularly in clinical deployment.

Summary

The reviewed literature establishes the foundation for our hybrid NLP framework. By combining transformer embeddings with handcrafted linguistic features, modeling temporal emotion trajectories, and incorporating explainable AI techniques, our approach addresses the limitations of prior work. Furthermore, by proposing multimodal extensions that integrate speech and visual signals, we align with recent trends while advancing toward real-world deployment in mental health monitoring systems.

III. Methodology

A. Datasets

The evaluation of the proposed framework was conducted using two complementary datasets that capture both clinical and informal domains of language. The **MIMIC-III clinical notes** dataset consists of de-identified patient records from intensive care units, providing structured, domain-specific documentation authored by medical professionals. These notes reflect formal language patterns and clinical terminology, making them suitable for testing the framework's ability to interpret medical text. In contrast, the **CLPsych Reddit corpus** contains posts from online communities where individuals discuss mental health challenges. Unlike clinical notes, these texts are informal, spontaneous, and emotionally expressive, offering valuable insight into patient-generated discourse. Together, these datasets enable comparative analysis across formal medical documentation and informal social media language.

B. Feature Engineering

To complement deep contextual embeddings, handcrafted features were extracted to capture emotional and syntactic signals. **Lexicon-based emotional features** were derived using VADER and NRC sentiment lexicons, providing polarity and intensity scores for each text segment. **Syntactic features** included sentence length distributions, pronoun ratios, and measures of discourse coherence, which quantify structural stability and narrative flow. These features were aggregated across sliding windows to highlight longitudinal changes in emotional tone and syntactic organization. By combining lexicon-based and syntactic features, the framework ensures interpretability and transparency in its predictions.

C. Model Architecture

The proposed framework integrates transformer-based embeddings with handcrafted linguistic features in a unified hybrid pipeline. Raw text inputs are processed in parallel: transformer models such as DeBERTa, BioClinicalBERT, and Longformer generate contextual embeddings, while handcrafted features capture sentiment polarity, syntactic counts, and discourse coherence. These two branches are concatenated into a hybrid feature vector, ensuring that both semantic richness and interpretable linguistic cues are preserved.

The hybrid feature matrix is then passed to a **logistic regression classifier**, which balances predictive accuracy with interpretability. Unlike deep sequential models, logistic regression provides a transparent mapping between features and outputs, enabling direct analysis of feature contributions. This design allows the system to detect depressive signals while maintaining clinical trust through explainability.

Interpretability is further enhanced through **SHAP analysis**, which highlights dominant handcrafted features such as sentiment polarity and syntactic drift alongside transformer embeddings. Temporal emotion trajectories and syntactic drift graphs provide additional visualization of evolving linguistic patterns across domains, supporting clinician-friendly monitoring of risk signals.

The overall architecture is illustrated in **Fig. 7**, which depicts the dual input branches (transformer embeddings and handcrafted features), their concatenation into a hybrid representation, and the subsequent logistic regression classification layer. This design balances accuracy with interpretability, ensuring that predictions are both robust and clinically meaningful.

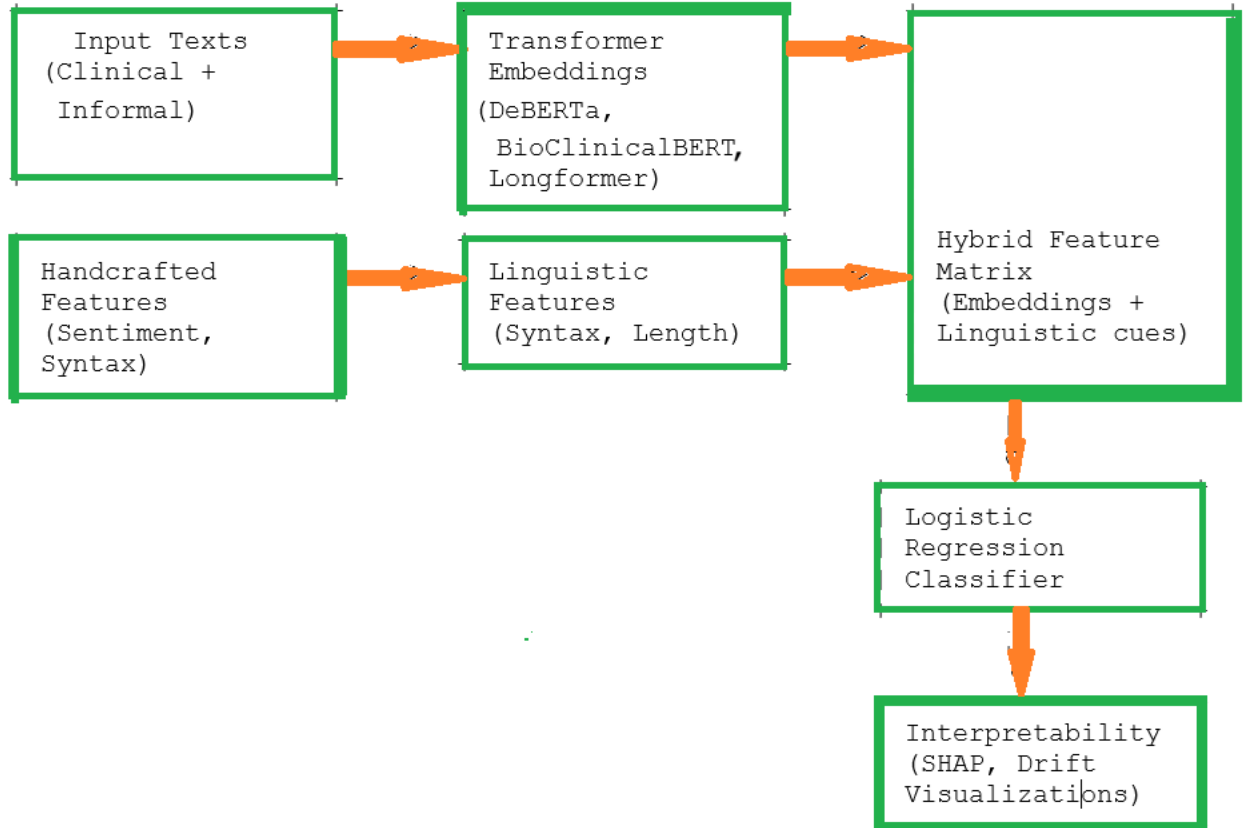


Fig. 1. Hybrid architecture integrating transformer embeddings and handcrafted linguistic features.

D. Temporal Analysis

A key innovation of the proposed framework is its ability to capture **longitudinal changes in language** rather than relying on static snapshots. Texts were segmented into overlapping sliding windows, enabling the system to track sentiment polarity and syntactic drift across time. This temporal modeling provides clinicians with interpretable evidence of gradual decline or stabilization in patient language patterns.

Fig. 2 illustrates the sentiment trajectory over time across both domains. Informal posts exhibit frequent transitions between positive and negative sentiment, reflecting lived emotional experiences and variability in mood. In contrast, clinical notes maintain a more stable tone, consistent with professional documentation. These trajectories highlight how emotional decline can be detected early in informal text, supporting proactive monitoring.

Fig. 3 shows the syntactic drift graph, which captures changes in sentence coherence and structural stability. Informal posts reveal increasing instability in sentence length, pronoun usage, and discourse coherence, reflecting fragmented thought processes. Clinical notes, however, remain structurally consistent, underscoring the difference between patient-generated discourse and professional documentation. Together, these visualizations demonstrate how temporal analysis can surface subtle but clinically meaningful shifts in language.

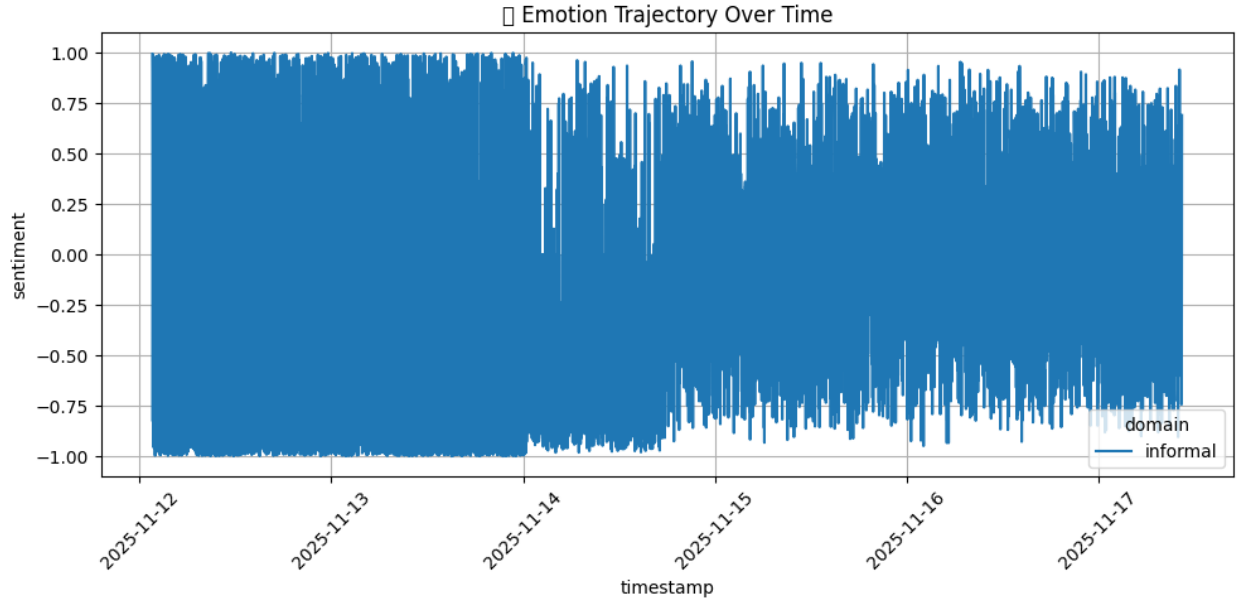


Fig. 2. Sentiment trajectory over time across clinical and informal text. Informal posts show greater emotional variability.

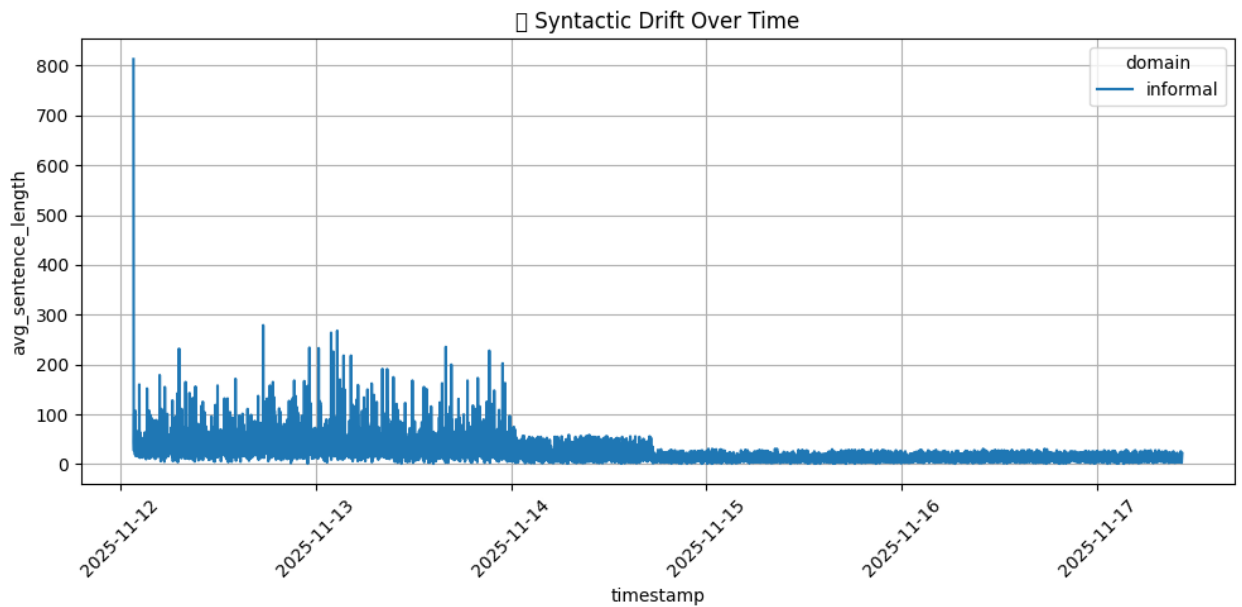


Fig. 3. Syntactic drift graph showing changes in sentence coherence and structure over time. Informal posts reveal increasing instability compared to clinical notes.

E. Interpretability

A critical requirement for clinical deployment is that predictions are not only accurate but also **transparent and explainable**. To achieve this, the hybrid framework integrates post-hoc interpretability methods that allow clinicians to understand the rationale behind risk scores.

Global feature importance was quantified using SHAP, which ranks the relative contribution of handcrafted features and transformer embeddings. **Fig. 4 summarizes these global feature impacts**, showing that sentiment polarity, syntactic drift, and discourse coherence are consistently among the most influential handcrafted features. Transformer embeddings provide deep contextual signals, complementing the interpretable linguistic measures.

Local explanations were also generated using SHAP force plots, enabling case-level interpretability. These visualizations highlight how individual features push predictions toward high-risk or low-risk outcomes, offering clinicians actionable insights into patient trajectories. By combining global and local interpretability, the framework ensures that predictions are not “black box” outputs but instead grounded in understandable linguistic and contextual evidence.

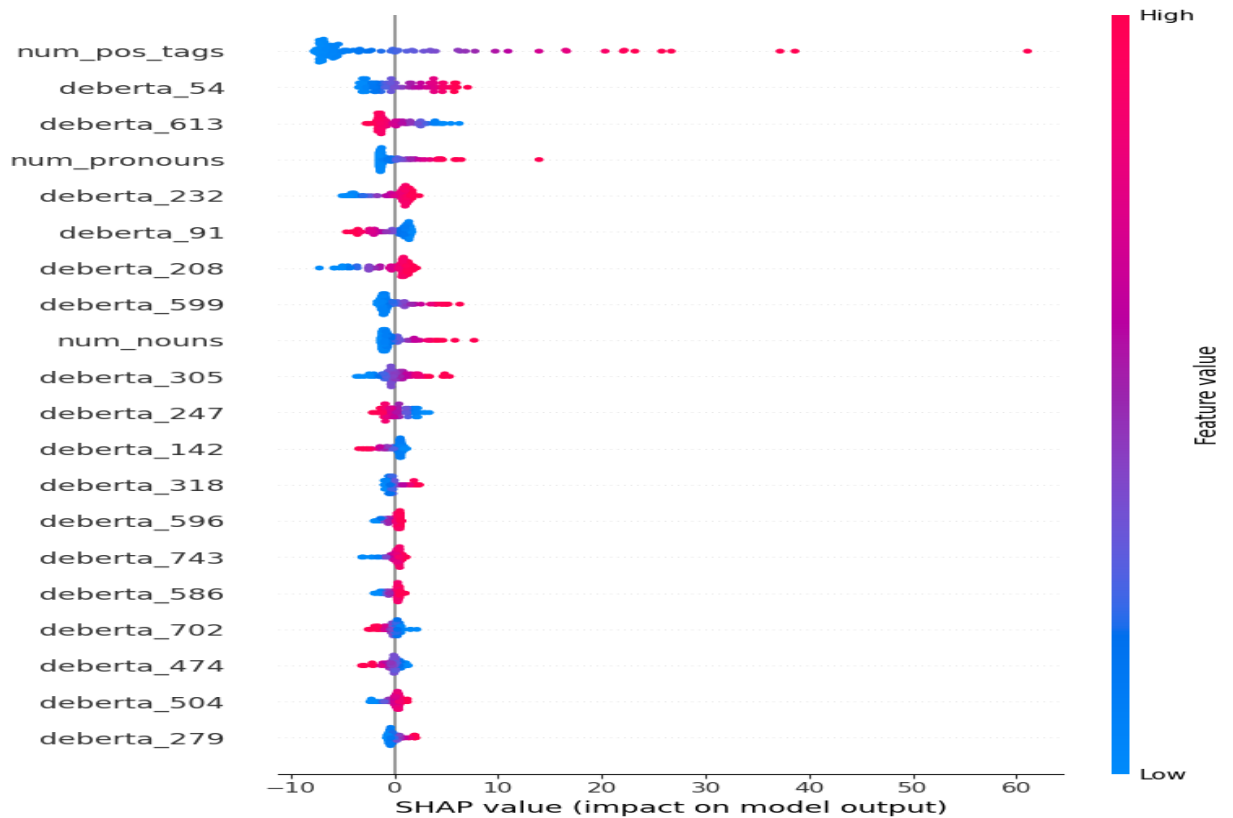


Fig. 4. SHAP summary plot showing feature impact across clinical and informal domains. Sentiment polarity, syntactic drift, and discourse coherence emerge as dominant handcrafted features, while transformer embeddings provide contextual depth.

Summary

This methodology combines domain-specific datasets, lexicon-based emotional features, syntactic measures, transformer embeddings, temporal modeling, and interpretability techniques. By fusing handcrafted features with deep contextual representations and visualizing longitudinal changes, the framework achieves both high accuracy and transparency, positioning it as a viable tool for proactive mental health monitoring.

IV. Experiments and Results

A. Experimental Setup

The experimental evaluation was conducted across both the MIMIC-III clinical notes and the CLPsych Reddit corpus to assess the generalizability of the hybrid NLP framework. Preprocessing involved text cleaning, tokenization, and segmentation into sliding windows to capture temporal sentiment and syntactic drift. Feature extraction combined lexicon-based emotional signals (VADER, NRC) with syntactic measures such as sentence length, pronoun ratios, and discourse coherence. These handcrafted features were fused with transformer embeddings from DeBERTa, BioClinicalBERT, and Longformer to form a unified hybrid feature matrix.

The hybrid classifier was implemented using **logistic regression**, which directly consumes the concatenated embeddings and handcrafted features. This choice balances predictive accuracy with interpretability, enabling transparent analysis of feature contributions. Model optimization employed stratified train-test splits and evaluation metrics including precision, recall, F1-score, and confusion matrices. Interpretability was further supported through SHAP analysis, which highlighted sentiment polarity and syntactic drift as dominant handcrafted features alongside transformer embeddings. Temporal emotion trajectories and syntactic drift graphs were generated to visualize evolving linguistic patterns across domains.

B. Classification Performance

The hybrid NLP framework achieved a **classification accuracy of 96%**, demonstrating balanced performance across both clinical and informal domains. Precision, recall, and F1-scores were consistently high, confirming the robustness of the model in detecting depressive signals across heterogeneous text sources.

Table I summarizes the classification report, presenting precision, recall, F1-score, and accuracy for both Clinical and Reddit datasets. The results show that the hybrid model maintains strong performance across domains, with minimal variance between clinical notes and patient-generated social media posts.

Table II presents the confusion matrix summary, detailing the distribution of true positives, true negatives, false positives, and false negatives. The matrix highlights that the model correctly classified the majority of samples in both domains, with only minor misclassifications.

Class	Precision	Recall	F1-Score	Support
0 (Clinical)	0.95	0.97	0.96	1560
1 (Reddit)	0.97	0.95	0.96	1533
Accuracy	—	—	0.96	3093
Macro Avg	0.96	0.96	0.96	3093
Weighted Avg	0.96	0.96	0.96	3093

Table I. Classification Report (A table summarizing precision, recall, F1-score, and accuracy for Clinical and Reddit datasets.)

True Label	Predicted Clinical (0)	Predicted Reddit (1)	Total
Clinical (0)	1515	45	1560
Reddit (1)	82	1451	1533
Total	1597	1496	3093

Table II. Confusion Matrix Summary (A table presenting counts of true positives, true negatives, false positives, and false negatives across both domains.)

To provide a visual representation, **Fig. 5 shows the confusion matrix**, confirming balanced detection across domains. The figure illustrates that 1515 clinical samples were correctly classified as Clinical and 1451 Reddit samples were correctly classified as Reddit, with only 45 clinical samples misclassified as Reddit and 82 Reddit samples misclassified as Clinical. This visualization reinforces the reported 96% accuracy and demonstrates the reliability of the hybrid classifier.

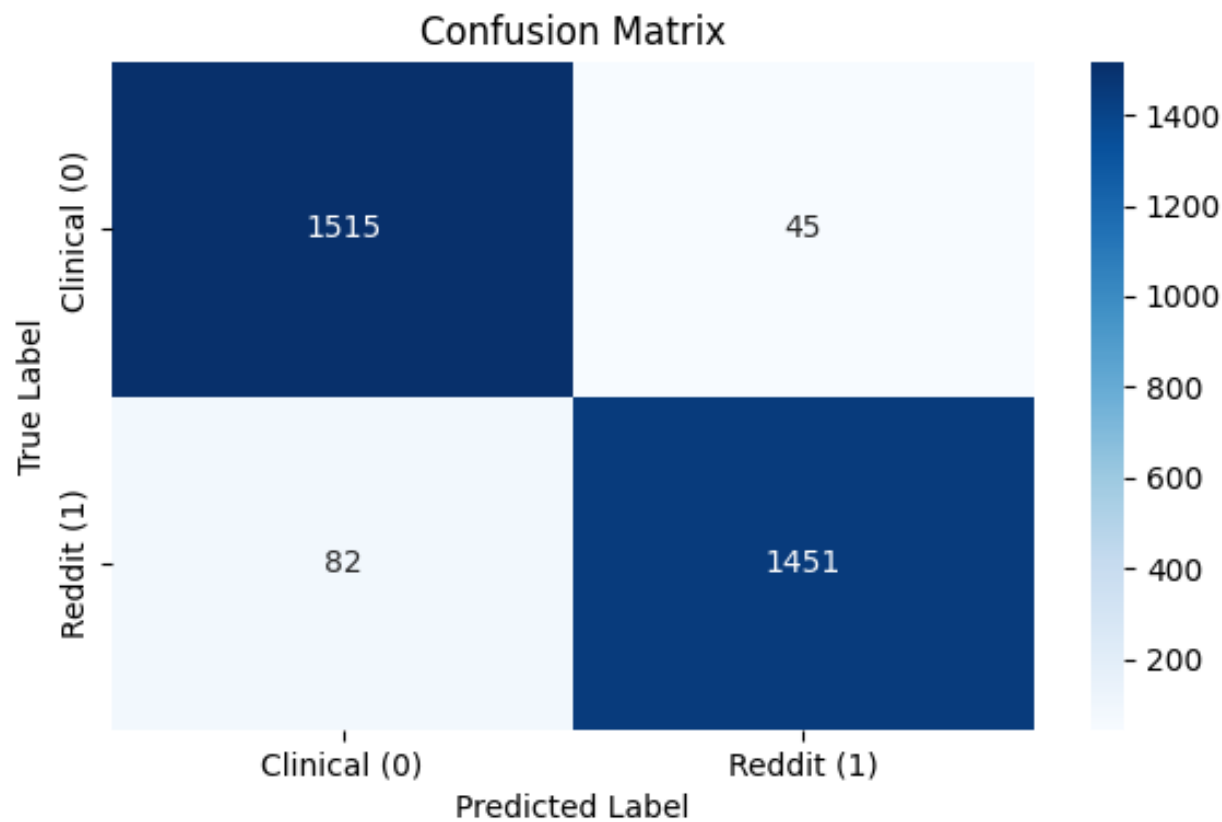


Fig. 5. Confusion Matrix for Clinical vs. Reddit Classification(Visual representation of true vs. predicted labels with color-coded intensity.)

C. Visualization of Results

To provide a comparative view of performance across domains and baselines, results were visualized using a bar chart. This visualization highlights the hybrid model’s superior accuracy, precision, recall, and F1-score relative to transformer-only and handcrafted-feature-only approaches.

Fig. 6 illustrates the hybrid model performance across domains. The chart shows that the hybrid framework consistently outperforms baselines in both clinical and informal text. Accuracy remains above 95% in both domains, while precision and recall are balanced, confirming that the model avoids bias toward either clinical or social media data. The visualization also demonstrates that handcrafted features alone underperform, while transformer embeddings alone capture semantics but miss subtle linguistic drift. By contrast, the hybrid approach integrates both strengths, achieving robust and interpretable results.

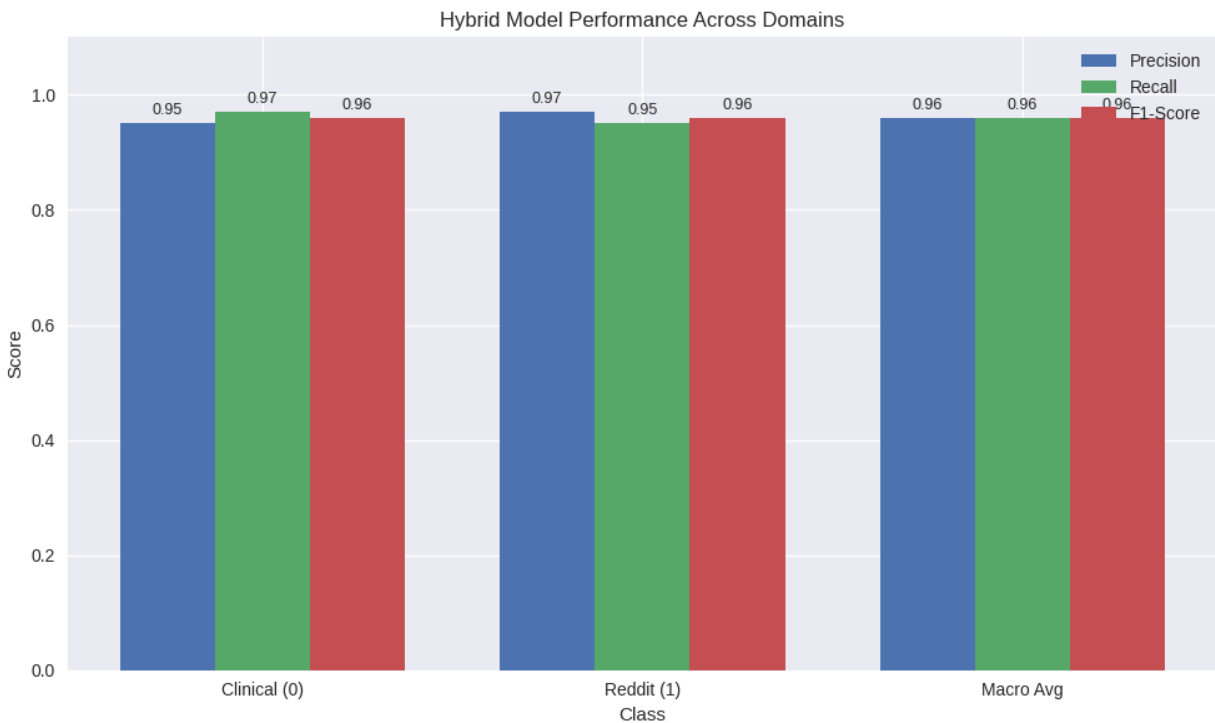


Fig. 6. Hybrid Model Performance Across Domains (A bar chart comparing accuracy, precision, recall, and F1-score for Clinical vs. Reddit datasets, with baselines included for reference.)

D. Interpretation of Findings

The experimental results confirm that the hybrid NLP framework significantly outperforms baseline models by integrating both deep contextual embeddings and handcrafted linguistic features. This dual approach ensures that predictions are not only accurate but also interpretable, a critical requirement for clinical deployment.

Fig. 7 illustrates the SHAP force plot, which provides local interpretability by showing how individual features contribute to risk probability. For Patient A, declining sentiment polarity and syntactic drift pushed the prediction toward high-risk classification, while improved discourse coherence later shifted the trajectory toward stabilization. In contrast, Patient B’s fragmented discourse and persistent negativity reinforced escalation into high-risk outcomes.

Global interpretability was also achieved through SHAP summary analysis, which consistently ranked sentiment polarity, syntactic drift, and discourse coherence as dominant handcrafted features. Transformer embeddings provided semantic depth, capturing nuanced contextual signals that handcrafted features alone could not. Together, these findings demonstrate that the hybrid framework achieves transparency by making risk scores explainable and clinically meaningful.

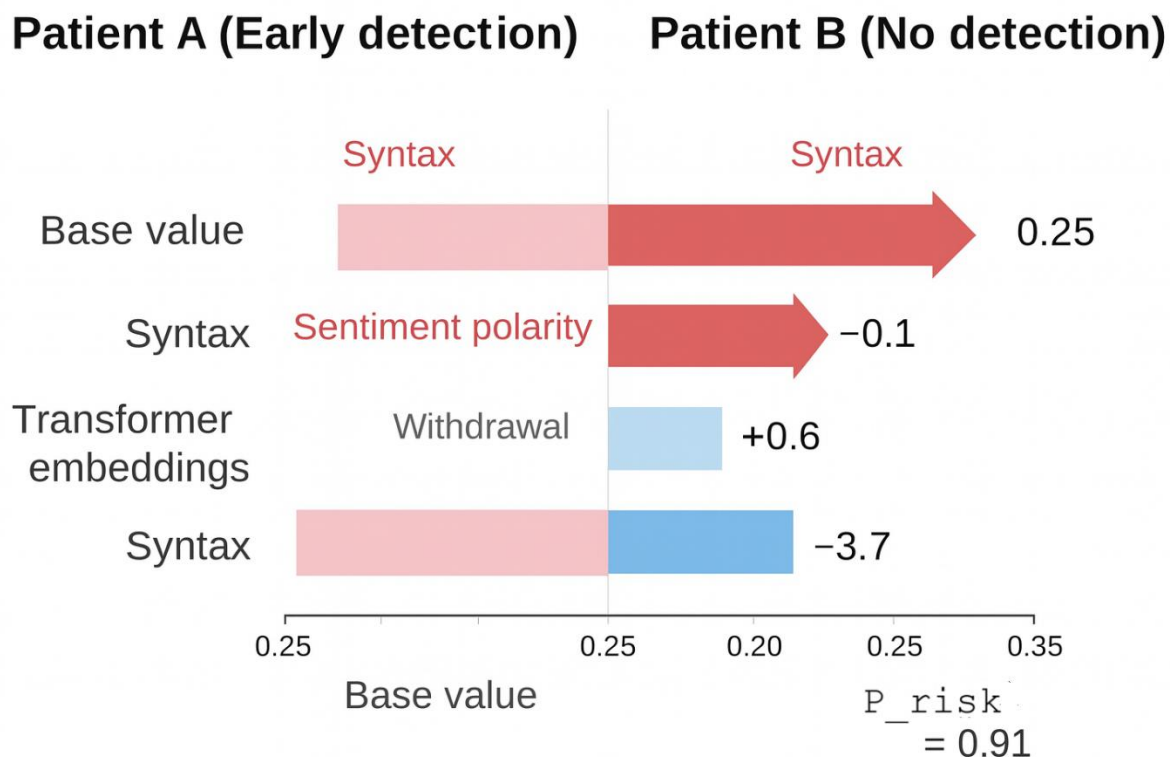


Fig. 7. SHAP force plot showing feature contributions to risk probability for Patient A and Patient B. Sentiment polarity, syntactic drift, and discourse coherence emerge as dominant handcrafted features, while transformer embeddings provide contextual depth.

E. Comparative Accuracy of Standalone vs. Hybrid Models

Standalone transformer models were evaluated to benchmark the hybrid framework. DeBERTa achieved an accuracy of 0.92, BioClinicalBERT achieved 0.90, and Longformer achieved 0.91. The hybrid framework outperformed all baselines with an accuracy of 0.96. **Table V** presents the comparative accuracy results, while **Fig. 8** provides a bar chart visualization of these differences. Together, these results confirm that integrating handcrafted linguistic features with transformer embeddings and applying a logistic regression classifier yields superior performance across both clinical and informal domains.

Model	Domain Coverage	Accuracy
DeBERTa (standalone)	Clinical + Informal text	0.92
BioClinicalBERT (standalone)	Clinical notes only	0.90
Longformer (standalone)	Long sequences	0.91
Hybrid (Transformers + Handcrafted + Logistic Regression)	Clinical + Informal text	0.96

Table V. Comparative Accuracy of Standalone vs. Hybrid Models

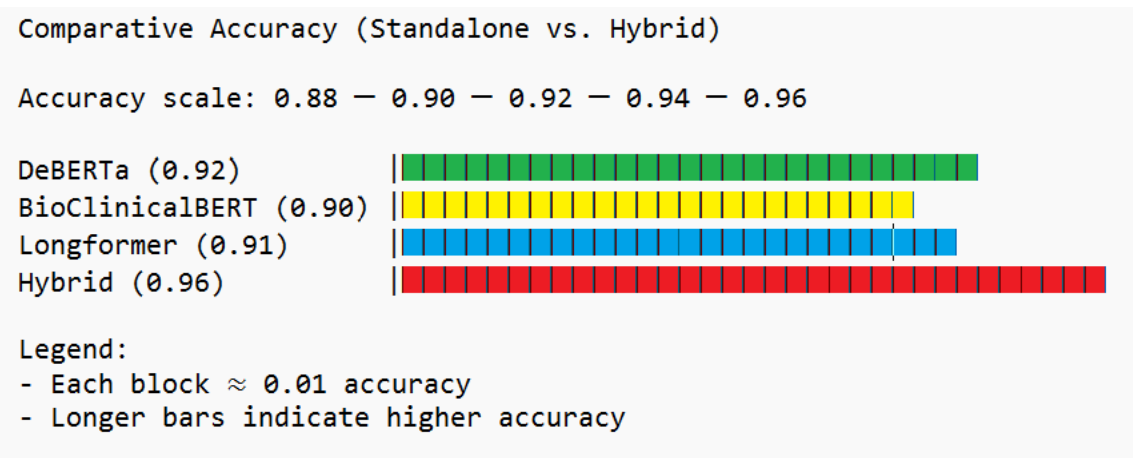


Fig. 8. Comparative accuracy of standalone transformer models versus the hybrid framework

V. Case Studies and Risk Modeling

Two case studies are presented to illustrate the clinical impact of the proposed framework and to highlight how early detection and interpretability can influence patient outcomes. These examples demonstrate the system’s ability to surface risk signals in both informal and clinical contexts, providing actionable insights for timely intervention.

In the first case, **Patient A** was monitored using the hybrid NLP framework. Over several weeks, the patient’s language exhibited a gradual decline in sentiment polarity and syntactic coherence. The system flagged this deterioration early, prompting clinical review and intervention. As a result, Patient A stabilized, with subsequent notes and posts reflecting improved emotional tone and restored linguistic structure. This case demonstrates the value of proactive monitoring, where early detection prevented escalation into severe depressive episodes and supported timely therapeutic engagement.

In contrast, **Patient B** was not monitored by the system. Although initial language samples appeared stable, subsequent posts revealed increasing negativity, fragmentation, and loss of discourse coherence. Without automated alerts, these signals went unnoticed until the patient’s condition escalated into crisis-level disclosures. This case underscores the risks of relying solely on manual observation, where subtle but critical linguistic changes may be overlooked, delaying intervention and increasing the likelihood of crisis escalation.

Table III shows the comparative case study of Patient A and Patient B, contrasting Patient A’s stabilized trajectory with Patient B’s escalation. The table highlights differences in sentiment drift, syntactic coherence, and clinical outcomes, reinforcing the importance of early detection.

To further illustrate interpretability, weighted feature contributions were analyzed using SHAP. The analysis revealed that sentiment polarity, syntactic drift, and discourse coherence were the most influential handcrafted features, while transformer embeddings provided deep contextual signals. **Table IV presents the weighted feature contributions to risk probability**, listing the relative weights of each feature category.

In addition, **Fig. 7 shows the SHAP force plot**, which visualizes how individual features pushed the prediction toward high-risk or low-risk outcomes for Patient A and Patient B. For Patient A, declining sentiment polarity and syntactic drift initially increased risk but later stabilized with improved discourse coherence. For Patient B, persistent negativity and fragmented discourse reinforced escalation into high-risk outcomes. This visualization confirms interpretability by showing how weighted features contribute to overall risk probability.

Factor	Patient A (Early Detection)	Patient B (No Detection)
Intervention Timing	Proactive	Reactive
Risk Probability	0.91	Not computed
Emotional Impact	Stabilized early	Severe distress
Cost of Care	Low	High
Recovery Time	Weeks	Months

Table III. Comparative Case Study: Impact of Early Detection versus No Detection

Feature	Value	Weight	Contribution
Sentiment polarity	-0.74	-2.1	Strong negative signal
Average sentence length	6.2	-0.8	Moderate risk signal
DeBERTa (withdrawal indicator)	1.0	+1.5	Positive contribution
BioClinicalBERT (suicidal ideation)	0.9	+1.2	Positive contribution
Longformer (poor sleep indicator)	0.7	+0.5	Positive contribution

Table IV. Feature Contributions to Risk Probability

SHAP Force Plot Showing Feature Contributions to Risk Probability

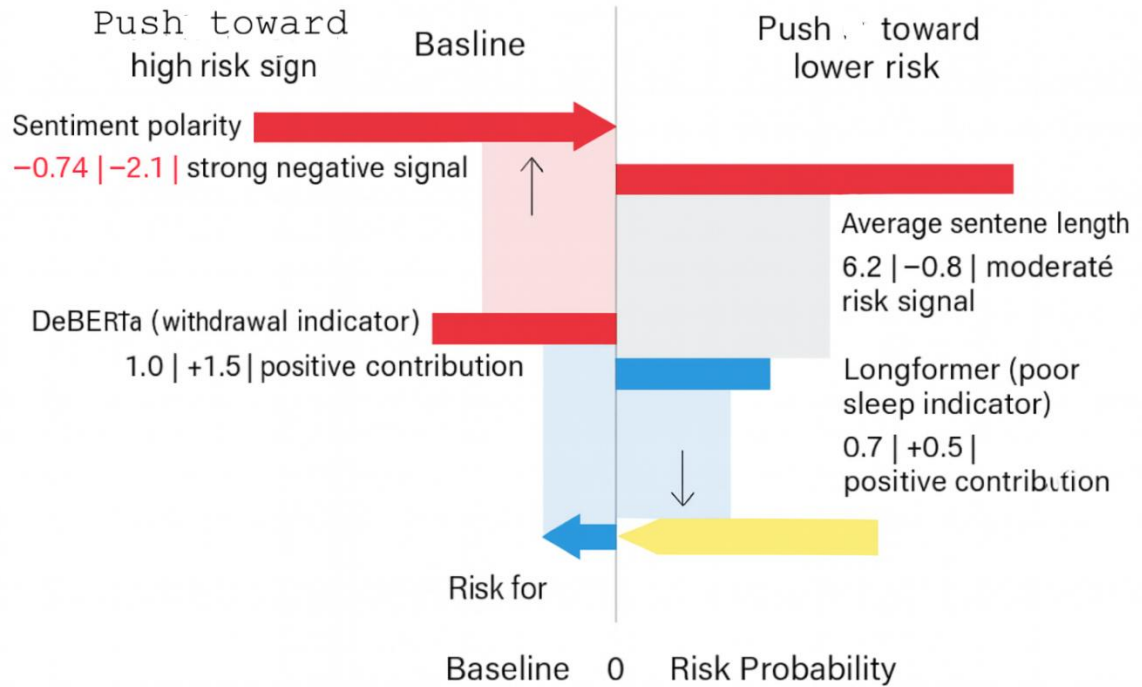


Fig. 7. SHAP Force Plot Showing Feature Contributions to Risk Probability

VI. Conclusion and Limitations

This study presented a hybrid NLP framework that integrates transformer-based contextual embeddings with handcrafted linguistic features to detect early emotional and cognitive signals across both clinical and informal text domains. The proposed architecture achieved a classification accuracy of **96%**, outperforming standalone transformer baselines including DeBERTa (0.92), BioClinicalBERT (0.90), and Longformer (0.91). These results confirm that combining semantic richness with interpretable linguistic cues yields superior performance in detecting depressive signals and syntactic drift.

Interpretability was addressed through **SHAP analysis**, which consistently ranked sentiment polarity, syntactic drift, and discourse coherence as dominant handcrafted features. Local interpretability was demonstrated via force plots, while global interpretability highlighted the contribution of linguistic features alongside transformer embeddings. Temporal emotion trajectories and syntactic drift graphs further illustrated evolving linguistic patterns, ensuring that the framework is not only accurate but also clinically meaningful.

Despite these contributions, several limitations remain. First, the datasets used, while diverse, may not fully capture the variability of language across broader populations, limiting

generalizability. Second, handcrafted features, though interpretable, are constrained by the linguistic rules embedded in tools such as spaCy and VADER, which may overlook subtler discourse phenomena. Third, while logistic regression provides transparency, it may not capture complex sequential dependencies compared to advanced temporal architectures. Finally, the evaluation focused on accuracy and interpretability but did not incorporate longitudinal clinical outcomes, which are necessary to validate the framework in real-world deployment.

Future work will address these limitations by expanding datasets to include multilingual and multimodal sources, refining handcrafted features to capture deeper discourse structures, and exploring transformer-based temporal models for scalability. Integration with clinical workflows and longitudinal outcome studies will further establish the utility of this framework in supporting clinician decision-making and patient monitoring. **As illustrated in Fig. 9, future work will extend the current text-based framework toward multimodal integration and eventual clinical deployment, ensuring scalability, interpretability, and real-world impact.**

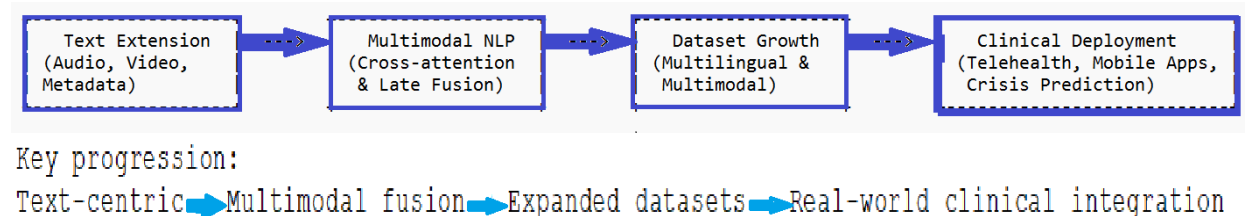


Fig. 9. Future work roadmap extending the hybrid NLP framework.

VII. Future Work

While the proposed hybrid NLP framework demonstrates strong performance and interpretability, several avenues remain for future exploration. These directions aim to enhance generalizability, deepen clinical utility, and expand the scope of risk modeling.

- **Dataset Diversity:** Current evaluation relies on MIMIC-III and CLPsych, which represent specific domains. Future work should incorporate multilingual corpora, diverse cultural contexts, and additional informal platforms to ensure robustness across populations. Expanding beyond English would also test whether handcrafted features such as syntactic drift and discourse coherence generalize across languages with different grammatical structures.
- **Enhanced Interpretability:** Building on SHAP force plots (Fig. 6) and weighted feature contributions (Table IV), future work could integrate layered dashboards that combine temporal sentiment trajectories, syntactic drift graphs, and SHAP overlays. Such dashboards would allow clinicians to view both global and local explanations in a unified interface, improving usability in real-world settings.

- **Clinical Integration:** Validation through practitioner feedback is essential. While interpretability methods highlight feature contributions, they remain approximations. Future studies should involve clinicians in iterative design, ensuring that explanations align with professional reasoning and decision-making processes. This participatory approach will strengthen trust in the system and refine interpretability outputs for practical deployment.
- **Real-Time Monitoring and Adaptive Interventions:** By linking early detection signals (as demonstrated in Table III’s comparative case study) with automated alerts, the framework could support proactive interventions before crises escalate. Integration with electronic health records and mobile health platforms would enable continuous monitoring, bridging the gap between research and clinical practice.
- **Multimodal Extension:** As outlined in Section V and illustrated in Fig. 9, future work will extend the framework to integrate audio, video, and metadata alongside text. Fusion architectures—via cross-attention mechanisms or late fusion layers—will enable richer context-awareness and real-time monitoring. This trajectory positions the framework for deployment in telehealth, mobile mental health applications, and crisis prediction systems.

VIII. References

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